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REVIEW

The impact of targeted physical activity interventions on inhibition control in children and adolescents with ADHD: a meta-analysis

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Abstract

This meta-analysis aims to identify intervention parameters (e.g., intensity, session time, frequency, and intervention duration) of targeted physical activity (PA) to improve inhibition control (IC) in children and adolescents with ADHD. The systematic search was conducted across PubMed, Web of Science, Embase, and Cochrane Library. In this meta-analysis we assessed the overall effect of PA interventions and then conducted subgroup meta-analyses and meta-regression to explore potential sources of heterogeneity. Finally, we checked publication biases and performed sensitivity analyses. Sixteen studies (591 participants aged 7–17 years) were included. The result showed that targeted PA interventions improved IC with high heterogeneity (SMD = 0.71, 95% CI [0.30, 1.11], $I^2 = 83.11\%$). Long-term interventions improved IC (SMD = 0.84, 95% CI [0.36, 1.31], $p < 0.01$), whereas acute interventions did not. Although no statistically significant group difference in long-term interventions was found ($p > 0.05$), examination of individual studies suggested greater improvements with session time around 60 min (SMD = 1.43), 2–3 times/week (SMD = 1.04), and intervention duration of at least 8 weeks (≥8 weeks: SMD = 0.49; ≥12 weeks: SMD = 1.19). Meta-regression revealed an inverted U-shaped relationship between frequency and intervention effects ($\beta_{frequency} = 2.15$; $\beta_{frequency*frequency} = -0.57$). Targeted PA interventions could improve IC in children and adolescents with ADHD, particularly designed with a moderate frequency. Although session time and intervention duration in long-term interventions showed no statistical significance in this meta-analysis, examination of individual studies revealed their promising results. Further investigations are therefore recommended.

Keywords Physical activity · Children · Adolescents · ADHD · Inhibition control

Introduction

Inhibition control (IC) is a core component of executive function that suppresses or regulates attention and behavior in the presence of inappropriate stimuli or distractions. As a key component of cognitive and socioemotional development [1], IC supports self-regulation, including emotional control, impulse inhibition, and adherence to social norms [2]. Developmentally, IC is fundamental to brain maturation and cognitive growth during childhood and adolescence, reflecting protracted development of the prefrontal cortex and supporting higher-order functions, such as planning, problem solving, goal maintenance, and complex emotion regulation [2, 3]. Specifically, during middle childhood IC shows rapid improvements as the prefrontal cortex matures [4]. In adolescence, IC may reach near-adult levels but remains sensitive to emotional and social influences by imbalance between emotion-related neural systems and still-developing cognitive control network, rendering IC less stable [5]. Psychosocially, there is a relationship between greater IC and better classroom self-regulation, improved peer relationships, and fewer behavioral problems [6]. Educationally, IC is positively associated with classroom learning, sustained attention, and academic performance [2, 7]. Conversely, deficits in IC could lead to learning difficulties, poor academic performance, and emotional problems. These deficits are more pronounced in children and adolescents with attention deficit hyperactivity disorder (ADHD) due to its key characteristics such as inattention, hyperactivity, and impulsivity [2, 8]. Untreated IC deficits during childhood

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often persist into adulthood and have a consistent, negative impact on academic achievement and social interaction [9].

Current treatments of IC deficits rely on pharmacological approaches [10]. These approaches, however, target primarily core symptoms of inattention and hyperactivity rather than improving IC [11, 12]. Also, the benefits of medications are not sustainable in the absence of prescriptions. What's worse, long-term use of medication therapies (e.g., amphetamines or atomoxetine/stimulant) may elicit or exacerbate psychotic symptoms in children and adolescents with ADHD [13]. Concerns about pharmacological treatments' side effects, long-term reliance, and limited improvement in IC have spurred interests in alternative or complementary methods [11].

Physical activity as an alternative treatment

In response to limitations of pharmacological treatments, physical activity (PA), particularly targeted PA, has emerged as a non-medical and cost-effective intervention for improving IC in children and adolescents with ADHD [14]. In this review, targeted PA refers to physical activities that are intentionally designed to improve executive function. Engaging in targeted PA (e.g., aerobic activities, ball games) may enhance the function of frontal brain, contributing to the improvement of IC [15]. These cognitive benefits are mainly associated with sufficient nerve stimuli through release of brain derived neurotrophic factors (BDNF), and an increased level of cerebral blood flow [16]. When engaging in PA, children and adolescents with ADHD are required to show coordination, sustained effort, and motor planning, which help them maintain focus, practice self-control, and channel hyperactivity in proper physical movements [17]. Activities, such as soccer, may enhance IC by activating the prefrontal cortex through repeated demands for participation and response inhibition in dynamic and unpredictable environments [18].

To examine the effect of PA on IC in children and adolescents with ADHD, researchers have compared different PA types (e.g., open-skill vs. close-skill, aerobic activity vs. cognitively engaging activity) [14, 19–21]. In this text, open-skill activities involve movements in unpredictable and dynamic environments (e.g., table tennis), and closed-skill activities are performed in a relatively stable and predictable environment (e.g., running) [19]. Aerobic activities refer to sustainable, rhythmic, and continuous exercises (e.g., swimming) that elevate the heart rate and breathing by mainly engaging large muscle groups [22]. Cognitively engaging activities (e.g., exergaming) are goal-directed and characterized with cognitive demands such as attention flexibility, inhibition control, planning [23]. However, meta-analysis studies focusing on children and adolescents with ADHD revealed that, regardless of types, targeted PA interventions generally yielded moderate improvements in executive function, including IC, and the effectiveness did not significantly differ across activity types [14, 21, 24]. For example, Song and his collogues found both open-skill and close-skill PA improved IC but no significant differences between the two groups [14]. Likewise, Liang reported there was no group difference between aerobic activity and cognitively engaging activity [21]. These results post a challenge for scholars and practitioners to pinpoint "best practices" to improve IC in children and adolescents with ADHD. Thus, it may be more meaningful to examine common intervention parameters that underline all types of physical activity, such as intensity, session time, frequency, and intervention duration.

Physical activity intervention parameters

Regarding the four intervention parameters, intensity refers to the level of physiological load achieved by participants during PA, usually expressed as a percentage of maximum heart rate (HRmax). Session time is defined as the average duration of PA for each session, measured in minutes (min). Frequency is the number of times by which a participant engages in PA per week. Intervention duration is the length of time from the beginning to the end of the entire intervention, usually in weeks.

Among these parameters, intensity has received most attention in research. From a neurocognitive perspective, researchers found moderate-intensity PA improved IC by promoting dopamine release and prefrontal cortex activation without depleting attentional resources [12]. This effect can be seen in meta-analysis studies that show moderate-to-vigorous intensity PA yielded optimal results for IC in ADHD participants [25, 26]. Researchers, however, found excessive intensity may cause cognitive fatigue and therefore limit IC improvement [27]. These findings indicate that optimal IC task performance occurred during PA at a moderate intensity level [28]. Moreover, emerging neurophysiological evidence suggests that excessively high intensity PA may lead to downregulation of prefrontal cortex activation and induce negative affective responses (e.g., discomfort and frustration), which can consequently impair executive functioning [29]. Interestingly, PA with low intensity, such as Yoga and Tai Chi, also showed positive effects on IC, likely due to their high cognitive demands rather than physical exertion [30, 31]. Despite the growing body of literature focusing on intensity, other intervention parameters including session time, frequency, and duration, have often been overlooked. Therefore, this meta-analysis will not only examine the effect of intensity based on previous studies,

but also explore the dose-response relationship between these parameters and the effectiveness of PA interventions.

Session time and frequency determine the balance between stimulation and recovery. A 60-min per session is considered as the effective setting to balance the neural stimulation and fatigue [23, 32]. Shorter session times (e.g., < 30 min) may not yield meaningful improvement in IC, while extended session times (e.g., >90 min) may result in diminishing cognitive benefits due to mental fatigue [27, 33]. For frequency, 1–3 sessions per week appear to be most effective due to neuroplastic adaption cycles [2, 34, 35]. Although researchers reported cognitive benefits might be associated with higher frequencies [36], these benefits could be offset when combining with high intensity or longer session time, largely due to the reduction in compliance and cognitive resource exhaustion [33]. Thus, a moderate frequency combined with proper intensity and optimal session time may provide a sustainable improvement on IC in children and adolescents with ADHD.

For intervention duration, a growing body of studies suggest that long-term interventions were effective in accumulating cognitive benefits and sustaining measurable improvement [37–40]. Concerning time efficiency, researchers also explored the effectiveness of PA with a minimum duration. Some studies asserted that acute interventions (i.e., a single session of PA without repeated engagement over time) could enhance IC via short neuroplastic adaptions and behavioral improvement [12, 15, 41]. Other studies, however, argued that acute interventions might not yield significant effects or sustained improvement on IC [21, 42]. These conflicting findings highlight the need of identifying effective parameters within PA interventions to improve IC in children and adolescents with ADHD [43].

While the benefits of PA for behavior and cognition in children and adolescents with ADHD have been described in the literature, there are still gaps. First, although many studies assessed how intensity moderated the effectiveness of PA on IC, few have explored the effect of time parameters, i.e., session time, frequency, and duration. Concurrently, scholars examining children and adolescents with ADHD found that heterogeneity in intervention outcomes might be due to intervention parameters rather than PA types [21]. Meanwhile, WHO guidelines and scholars also emphasized the research gap in dose-response studies regarding the effect of intervention parameters in improving IC in children and adolescents with ADHD [2, 35, 44]. This lack of analysis on how intervention parameters bring improvement in IC may limit the development of effective and targeted intervention designs. These gaps, hence, necessitate the need of a systematic meta-analysis to synthesize existing evidence and identify dose-response relationships, translating fragmented findings into an optimal PA intervention design.

The present study

Given the aforementioned reasons, it is necessary to identify how these parameters affect the outcomes of PA interventions [43]. With this purpose, this meta-analysis will focus on two research questions: (1) Do targeted PA interventions lead to measurable improvements in IC among children and adolescents with ADHD? (2) How do frequency, intensity, session time, and intervention duration moderate the effectiveness of PA programs on IC respectively? By conducting the present study, we hope to provide a unique perspective about how each of the intervention parameters affects the overall effectiveness of an intervention. The results can be further used by researchers and practitioners to design effective PA interventions for improving IC in children and adolescents with ADHD.

Method

Search strategy

This meta-analysis was registered in the Prospective Systematic Evaluation Registry (PROSPERO, CRD42024597006) and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [45]. Search strategy was performed across four databases, including PubMed, Web of Science, Embase, and Cochrane Library, that focus on biomedical research and healthcare interventions. The initial search was conducted from October 16th to November 9th, 2024, and was last updated on January 13th, 2025. Figure 1 presents the procedure of the search.

We searched titles, abstracts, and subject headings by using medical terms (MeSH) and relevant tree words, such as physical activity, executive function, children, adolescent, and ADHD. The Boolean Operators (AND, OR) for keywords combination are presented in Appendix Table 5. We also used a handsearching method to ensure a complete capture of the literature. We reviewed previous relevant meta-analysis studies and cross-checked references that may meet the inclusion criteria.

Screening and study eligibility

The inclusion criteria based on the PICOS principle (P: Population or Patient; I: Intervention; C: Comparison; O: Outcome; S: Study design) included: (1) Participants were children and adolescents with ADHD of any subtypes aged > 5 years and < 18 years. (2) Participants were diagnosed with ADHD by a professional organization or

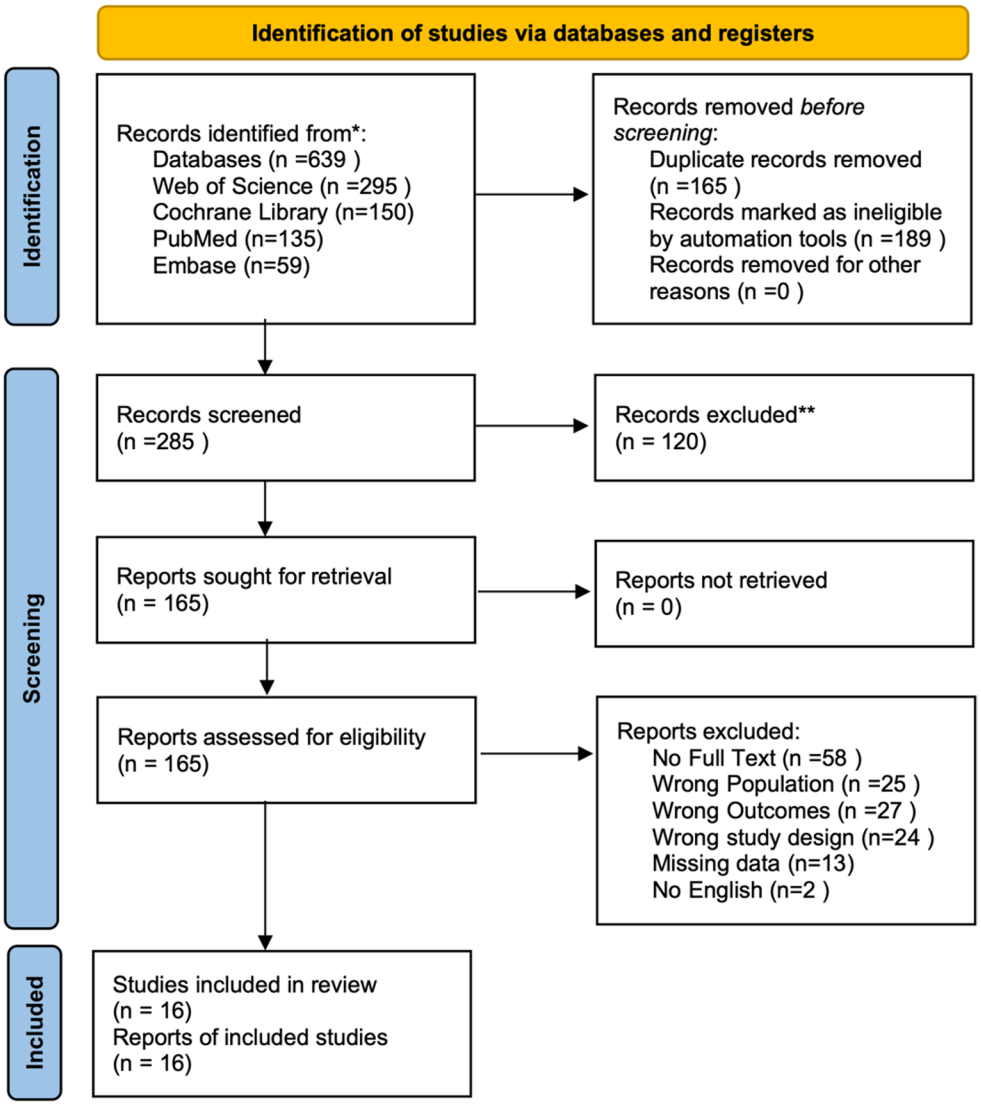


Fig. 1 The flow diagram of search process

validated rating scales, such as DSM-IV, DSM-V, ICD-10, SNAP-IV. (3) Studies must report PA interventions as the main approach to improve executive function outcomes, specifically IC, in children and adolescents with ADHD. (4) Studies must report at least one intervention parameter, including intensity, frequency, session time, or intervention duration. (5) Studies must assess and report inhibition control as an outcome, measured through validated behavioral or neuropsychological tasks (e.g., Stroop, Go/No-Go, or Flanker). (6) The study design was randomized controlled trial (RCT), quasi-experimental, or cross-over. (7) The study must be empirical, peer-reviewed, available in full text, and written in English.

The exclusion criteria were: (1) Participants aged ≤ 5 years or >18 years. (2) Participants were diagnosed with other mental disorders, such as autism, Tourette syndrome, epileptic disorders, or depression. (3) The study did not employ a PA intervention. (4) The study did not report IC outcomes. (5) The study was longitudinal or observational research, such as case studies, cross-sectional studies, or cohorts. (6) The study was a literature review, report, conferences paper, book chapter, or policy document. (7) The methodological quality was poor (Physiotherapy Evidence Database Scale [PEDro] score < 4) [46, 47]. The PEDro scale was chosen because it is a widely used and validated tool for evaluating RCT and quasi-experimental trials in physical activity and rehabilitation research. (8) The study was not written in English.

Study selection and data extraction

Based on the search of the four databases, all records were imported into Covidence, a web-based collaboration software platform designed to streamline the production of systematic and other literature reviews [48]. Duplicates and records identified as reviews were removed automatically

using Covidence. Two authors (Min and Chen) independently screened the title and abstract of the remaining studies according to the inclusion and exclusion criteria. During this stage, reviews, meta-analyses, and irrelevant studies—such as those not mentioned key terms related to PA, ADHD, or IC—were excluded. Full texts of potential eligible studies were then retrieved and assessed for eligibility. The final decision was made after screening the full text of included studies. In case of disagreement, there was a joint discussion and a consensus reached between the two authors (Min and Chen).

After screening the full text and assessing quality, the data were extracted, including information about: (1) first author, year of publication, and study area, (2) study design, (3) subject information (sample size, sex ratio, age), (4) physical activity intervention (type of intervention, exercise intensity, time of one session, frequency of physical activity, and total duration of intervention), (5) inhibition control measures and outcomes, including mean and standard deviations (SDs) of pre-test data and post-test data. If the study reported standard errors (SE) only, we converted SEs to SDs by using an established formula [49, 50]. Studies that did not report these values (e.g., mean, SD or SE, and sample size) were excluded. If multiple publications reported overlapping data from a same database, only the most recently published study was included. If there were two or more IC measurements, only the data with the most commonly used assessment was extracted. The study selection and data extraction process is presented in Fig. 1. Extracted outcome data (means, SDs, and sample sizes) were subsequently used to compute standardized mean differences (Hedges' g) with 95% confidence intervals, as detailed in the Data Analysis section.

Data coding and management

Intensity was coded using heart rate maximum (HRmax) categories, with reported and evaluated intensity < 60% HRmax considered low intensity activity (LPA), 60–75% HRmax as moderate intensity activity (MPA), >75% HRmax as vigorous intensity activity (VPA), and ≥ 60% HRmax as moderate-to-vigorous intensity activity (MVPA) [51]. When studies reported MVPA as a combined intensity category or reported a heart rate range (e.g., 70–80% HRmax) overlapping moderate and vigorous zones, the intensity was classified as MVPA. This coding strategy follows the PA guidelines used in previous meta-analysis studies [21, 35, 52]. Session time was coded into five categories: 30 min or less, [30 min, 45 min), [45 min, 60 min), [60 min, 90 min) and 90 min or longer [34, 44, 53]. Intervention frequency (times per week) was coded into three categories: 1 time, 2–3 times, and >3 times, based on the thresholds supported by previous meta-analysis studies and the WHO guidelines [35, 54].

Referred to previous systematic review and meta-analysis studies, an acute intervention was defined as a one-time, single-session intervention with no ongoing cycles [55], and a long-term intervention lasts at least four weeks [56–58]. To be noted, in the present study, all included acute interventions were single-session designs, while all long-term intervention lasted four weeks or longer. Particularly in this study, acute and long-term interventions differ fundamentally in how the four intervention parameters are included. An acute intervention contains a single session and therefore only includes two parameters: intensity and session time [15]. In contrast, long-term interventions are often repeated over multiple weeks, involving all four intervention parameters [57, 58]. Then, based on the characteristics of included studies and other experimental studies, the intervention duration was firstly coded as acute intervention and long-term intervention separately, and the long-term intervention was then categorized as ≥ 4 weeks [59], ≥ 8 weeks [34], and ≥ 12 weeks [55]. Accordingly, if heterogeneity exists, subgroup analysis for acute intervention was focused on the effect of intensity and session time on IC, and for long-term intervention we assessed the effect of all four parameters—intensity, session time, frequency, and intervention duration.

Assessment of the methodological quality of included studies

Quality of methodology of included articles was assessed using the Physiotherapy Evidence Database Quality Scale (PEDro scale). The PEDro scale assesses the clarity of participants eligibility, method of random allocation, blinding of subjects, assessors, and therapists, compliance rate, intention-to-treatment analysis, between-group statistical comparisons, as well as point estimates and variances. It includes 11 scales in four categories: "poor" (scores 0–4), "fair" (4–5), "good" (6–8), and "excellent" (9–10) [60]. Studies with a PEDro score < 4 were excluded, as such scores indicate a poor methodological quality and high risk of bias [46, 47]. The assessment was conducted by the first two authors. Disagreements were discussed until a consensus was reached.

Data analysis

Focusing on the research questions, we first used Stata 18.0 software to assess the overall effect of PA interventions on IC in children and adolescents with ADHD. All effect sizes were calculated using unadjusted means and standard

deviations reported in the studies (or converted from SEs) to ensure consistency across studies. We used the standardized mean difference (SMD, *Hedges' g*) to represent the effect size of the intervention, and reported 95% confidence intervals (CI) to assess the confidence of effect sizes [61]. The effect size was calculated using a random-effects model due to the nature of experimental studies [62–64]. SMD values < 0.2, from 0.2 to 0.5, from 0.5 to 0.8, ≥ 0.8 were considered as fair, small, moderate, and large effect sizes, respectively [61]. The Forest plot was used to present the result of meta-analysis. Heterogeneity was assessed using I^2. When $I^2 \leq$ 25%, it means low heterogeneity among studies, 25% < $I^2 \leq$ 50% means moderate heterogeneity, 50%< $I^2 \leq$ 75% means substantial heterogeneity, and I^2 >75% means high heterogeneity [65]. The Galbraith plot was used to illustrate the heterogeneity results.

Then, we conducted subgroup meta-analyses and meta-regression analyses to investigate potential moderators of intervention effects. These analyses were planned as part of the research aim to explore intervention parameters but were statistically conducted only when heterogeneity exceeded I^2 >50%. This practice is in line with best practices for meta-analytic methodology [50, 66]. If I^2 was larger than 50%, we conducted subgroup meta-analyses stratified by intervention durations (acute vs. long-term), as explained previously. Between-group heterogeneity was tested using Q_b, with $p < 0.05$ indicating statistically significant subgroup differences [64]. If significant between-group heterogeneity was detected, further subgroup meta-analyses were performed based on key intervention parameters [21, 50]. Specifically, for long-term interventions, subgroup analyses were conducted according to intensity, session time, frequency, and intervention duration. For acute interventions, intensity and session time were examined. A *p*-value less than 0.05 in these computations indicated a statistically significant moderating effect of the corresponding intervention parameter on IC. Although intervention intensity was pre-specified as a moderator of interest, subgroup meta-analysis or meta-regression for intensity was not conducted due to the limited number of studies reporting clearly defined intensity. Instead, we did a descriptive summary of effect sizes across studies categorized by intervention intensity (e.g., LPA, MPA, or MVPA) to explore its potential impacts on IC.

Next, we used *R* software with the *metafor* package to examine potential dose-response relationships through meta-regression using SMDs as the dependent variable and session time, frequency, and intervention duration as continuous predictors. Drawing on the cognitive load theory and neurocognitive response model for PA and executive function, both of which posit that cognitive resources are limited and both insufficient and excessive task demands may impair cognitive performance [21, 67, 68], we tested both linear and quadratic models. In the quadratic model, we included a second-degree term (i.e., frequency^2) to detect a potential inverted-U shaped relationship. The two models were compared using F-test to evaluate which one was a significant better fit to the data.

Finally, we used Egger's test to assess publication bias. When Egger's test was significant, we applied Duval and Tweedis's trim-and-fill method to estimate the number of potential missing studies and to adjust the overall effect size [69]. Additionally, the sensitivity analysis was conducted using the leave-one-out (LOO) method by removing one study at a time to evaluate the reliability of the findings [70, 71].

Results

Search results

A total of 639 articles were retrieved from four electronic databases: Web of Science (n = 295), Cochrane Library (n = 150), PubMed (n = 135), and Embase (n = 59). After removing duplicates (n = 165) and automatically excluded reviews (n = 189), 285 records remained for title and abstract screening. Of these, 120 records were excluded for not meeting the inclusion criteria (e.g., meta-analysis, participants with autism and/or medical treatment). The remaining 165 articles were eligible for full-text screening. During full-text screening, 58 articles were excluded due to inaccessibility of the full text [72, 73], 25 for ineligible populations (e.g., participant with Autism Spectrum Disorder) [74–77], 27 for irrelevant outcomes (e.g., motor function) [78, 79], 24 for unsuitable study designs (e.g., not RCT) [80, 81], 13 for missing data [82, 83], and 2 for non-English languages (e.g., Spanish) [84, 85]. The remaining studies passed the methodological quality threshold (PEDro >4), and no studies were excluded at this stage. Finally, 16 articles were included in the meta-analysis [39–42, 59, 86–96] (see Fig. 1).

Description of studies

The total 16 studies included 591 participants (75.13% male) aged 7–17 years. Among them, four studies were conducted in China, three in Iran, two in Tunisia, two in the United Kingdom, two in the United States, and one each in Italy, Korea, and Turkey. The characteristics of the studies are presented in Appendix Table 6,7,8,9.

For the intervention parameters, eight studies reported moderate to vigorous intensities at an average of 50% to 80% maximum heart rate (HRmax). Specifically, three interventions employed MPA, four applied MVPA, and nine did not report PA intensity. Each session for included

studies ranged from 30 to 120 min with one to five times every week. Three studies focused on acute interventions, and 13 studies examined long-term interventions. All acute interventions involved a single session of 30 min or less, with no repeated engagement. Regarding long-term interventions, six studies reported session time lasting between 45 and less than 60 min, four between 60 and less than 90 min, and three with session time of 90 min or longer. For intervention frequency, two studies allowed participants to do PA one time per week, nine assigned 2–3 times per week, and two required >3 times per week. The intervention duration ranged from 4 to 15 weeks. Except for one study that lasted for one and half years (72 weeks) [91], three interventions were run ≥ 4 weeks, three lasted ≥ 8 weeks, and seven continued ≥ 12 weeks.

Methodological quality assessment

Table 1 shows the quality assessment results for the 16 articles using the PEDro scale. The scores ranged from 5 to 8 (Mean = 6.1, SD = 1.06), indicating the quality of included studies was at least "good." Among them, 11 studies were of good quality (scores ≥ 6), accounting for 70.58% of the total studies, and five studies were fair (scores ≥ 5). Note that none of the studies had a PEDro score equal or below 4; thus, all included studies demonstrated an adequate methodological quality.

Overall intervention effect

Overall, across 16 studies, the targeted PA interventions had a moderate, positive effect on the improvement of IC in children and adolescents with ADHD (SMD = 0.71, 95% CI [0.30, 1.11], I^2 = 83.11%; see Fig. 2). As part of per-specified analysis plan, subgroup analyses were conducted for acute and long-term interventions, based on observed heterogeneity. As mentioned previously, intensity and frequency were used as moderators in subgroup analyses for acute interventions. For long-term interventions, we examined how intensity, session time, frequency, and intervention duration moderated the observed intervention effects.

Subgroup meta-analysis for acute and long-term interventions

There were three studies that examined the acute effect of interventions on IC, and 13 studies examined the long-term effect (see Fig. 2). While the acute interventions showed no statistically significant effect with a small heterogeneity (SMD = 0.18, 95% CI [−0.21, 0.57], I^2 = 0.00%), the long-term interventions had a positive, large effect with a large heterogeneity (SMD = 0.84, 95% CI [0.36, 1.31], I^2 = 84.96%). Since there was no significant heterogeneity in the acute interventions, the following analyses were mainly focused on the long-term interventions.

Subgroup meta-analysis by intervention parameters in long-term intervention

To identify sources of heterogeneity, we performed subgroup meta-analyses focusing on three intervention parameters—session time, frequency and duration. As explained earlier, we did not examine the moderating effect of intensity in the subgroup meta-analysis and meta-regression due to insufficient data, but provided a descriptive summary of its observed effect sizes below.

Descriptive summary of intervention intensity

In long-term interventions, four studies implemented MVPA [39, 40, 59, 92]. These studies yielded moderate to large effect (SMD ranging from 0.74 to 1.30). One study utilized MPA, and the result indicated a non-significant effect (SMD = 0.18, 95% CI [−0.52, 0.88]) [95] (Table 2).

Subgroup meta-analysis by session time, frequency, and intervention duration

Although the test for group difference did not find statistical significance across these moderators ($p > 0.05$), examination of individual studies (see Fig. 3) showed moderate to large effect sizes for interventions with session time between 60 and less than 90 min (SMD = 1.43, 95% CI [0.77, 2.09]), frequencies of 2–3 times per week (SMD = 1.15, 95% CI [0.51, 1.78], $p < 0.01$), and intervention duration of at least 8 weeks (≥ 8 weeks: SMD = 0.49, 95% CI [0.08, 0.91]; ≥ 12 weeks: SMD = 1.19, 95% CI [0.37, 2.01]). However, due to the limited and unequal number of included studies in each category, these findings should be interpreted with caution.

Meta-regression for time parameters in long-term intervention

To explore the dose-response relationship between standardized effect size and time parameters, including session time, frequency, and intervention duration, we fit a linear model and a non-linear (quadratic) model to the data. While both models demonstrated a statistical significance, the F-test comparing the two models indicated a significant improvement in the overall model fit after including quadratic terms (see Table 3). Therefore, the subsequent analyses were focused on the quadratic model.

Table 1 Assessment of the methodological quality based on PEDro

Study	Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Q 9	Q 10	Q 11	Score (Out of 10)	Quality
Chang 2014 [42]	Y	Y	N	Y	N	N	N	Y	Y	Y	Y	6	Good
Pan 2016 [86]	Y	Y	N	Y	N	N	N	Y	Y	Y	Y	6	Good
Walters 2024 [41]	Y	Y	N	Y	Y	N	Y	Y	Y	Y	Y	8	Good
Memamoghaddam 2016 [39]	Y	Y	N	Y	N	N	N	Y	Y	Y	Y	6	Good
Nejati 2021 [88]	Y	Y	N	N	N	N	N	Y	Y	Y	Y	5	Fair
Capodieci 2019 [89]	Y	N	N	N	H	Y	N	Y	Y	Y	Y	5	Fair
Wexler 2021 [90]	Y	Y	N	Y	N	Y	H	H	Y	Y	Y	6	Good
Kadri 2019 [91]	Y	Y	N	Y	H	N	N	Y	Y	Y	Y	6	Good
Hattabi 2019 [40]	Y	Y	N	Y	N	N	N	Y	Y	Y	Y	6	Good
Booth 2024 [92]	Y	Y	N	Y	N	N	N	H	Y	Y	Y	5	Fair
Nejati 2021 [88]	Y	Y	N	Y	N	N	N	Y	Y	Y	Y	6	Good
Ji 2023 [59]	Y	Y	N	Y	N	N	N	H	Y	Y	Y	5	Fair
Pan 2019 [87]	Y	Y	N	Y	N	N	N	Y	Y	Y	Y	6	Good
Kuo 2024 [94]	Y	N	N	Y	N	N	N	Y	Y	Y	Y	5	Fair
Durguta 2020 [95]	Y	Y	Y	Y	Y	N	N	Y	Y	Y	Y	8	Good
Smith 2020 [96]	Y	Y	Y	Y	N	N	Y	Y	Y	Y	Y	8	Good

Y low risks of bias; *N* unclear risks of bias; *H* high risks of bias. Score: low risks of bias = 1; unclear risks of bias = 0; high risks of bias = 0. Q1 score is not included in the final score out of 10. Fair: 4–5 scores; Good: 6–7 scores; Excellent: 9–10 scores. Q1: eligibility criteria of participants were specified; Q2: random allocation to groups; Q3: allocation was concealed; Q4: the baselines among the groups were similar; Q5: there was blinding of all subjects; Q6: all therapists were blinded; Q7: all assessors were blinded; Q8: for at least one key outcome, the consistency were no less than 85%; Q9: intention-to-treat analysis was conducted; Q10: at least one key outcome was reported; Q11: for at least one key outcome, both point estimate and variability were reported

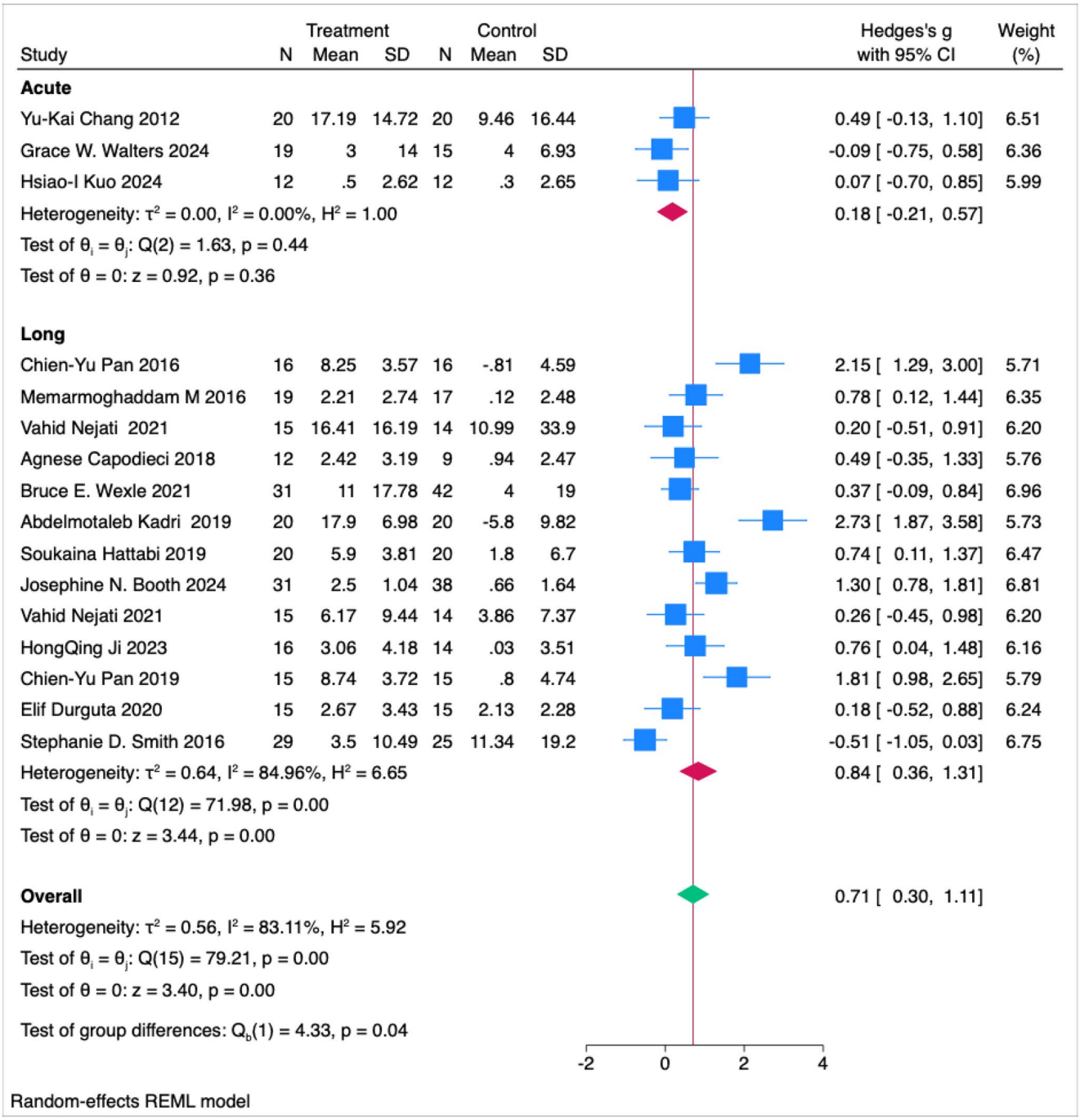


Fig. 2 Forest plot of subgroup analysis for the effect of acute interventions and long-term interventions. The left side shows the first author and year, sample size, mean, standard deviation of each study; the blue square represents the effect size; the horizontal line represents the 95% confidence interval; the red diamond represents the pooled effect size for each group; the green diamond represents the pooled effect size; the width of diamond represents 95% confidence interval. The right side shows Hedges's g and 95% confidence interval for each study

Table 2 Effect size of intensity for long-term interventions

Study	Hedges'g	95%CI	Intensity
Memarmoghaddam M 2016 [39]	0.78	[0.12, 1.44]	MVPA
Soubaina Hattabi 2019 [40]	0.74	[0.11, 1.37]	MVPA
Josephine N. Booth 2024 [92]	1.30	[0.78, 1.81]	MVPA
HongQing Ji 2023 [59]	0.76	[0.04, 1.48]	MVPA
Elif Durguta 2020 [95]	0.18	[−0.52, 0.88]	MPA

Table 4 shows that frequency and its squared term were statistically significant ($\beta_{frequency}$=2.15, p=0.04; $\beta_{frequency*frequency}$ = −0.57, p=0.03), indicating an inverted U-shaped relationship. Figure 4 shows the optimal effect may occur at around 2 times per week.

Table 3 Model fit indices and comparison

Model	DF	Adjusted R^2	F	p	Model Comparison
Linear Model	5	0.51	6.13	0.01**	F
Quadratic Model	8	0.75	8.44	0.01**	(3,9)=4.85 p=0.03*

Significance code: *p<0.05, **p<0.01; DF: degree of freedom; F-test vs. Linear: the difference test using F-tests for nested models

Results of publication bias and sensitivity analysis

The funnel plot (Fig. 5) shows an asymmetry. Egger's regression test indicated potential small-study effects (β=6.66, SE=3.10, t=2.15, p=0.05). After imputing two potential missing studies, the adjusted pooled effect size

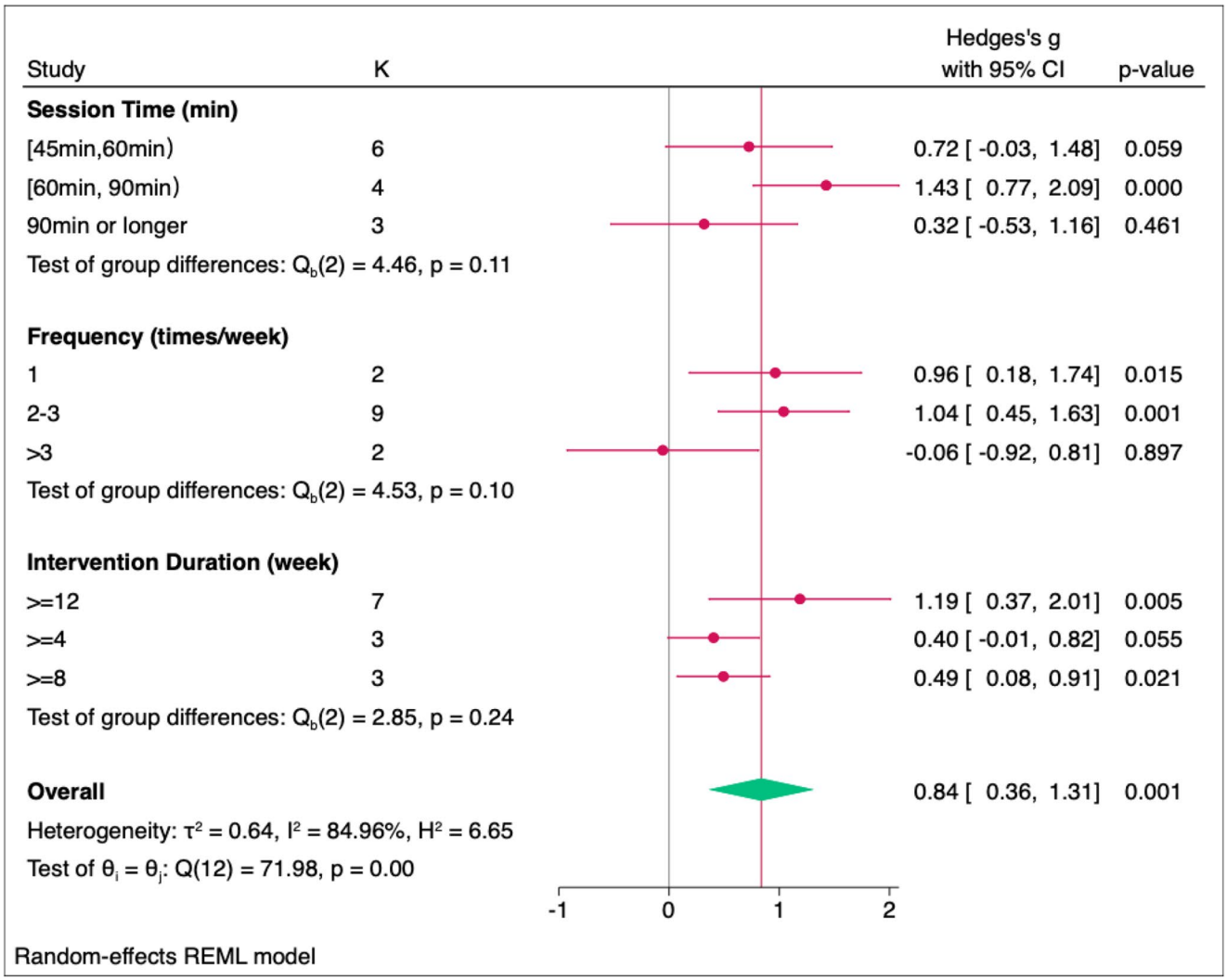


Fig. 3 Forest plot of subgroup analysis for long-term intervention based on session time, frequency, and duration. The left side shows subgroups based on intervention parameters and the result of Q_b statistic; [45min, 60min): between 45min and 60 min, excluding 60min; [60min, 90min): between 60min and 90 min, excluding 90 min; >= (≥): equal to or greater than; K: the number of included studies; the red horizontal line represents 95% confidence interval; the green diamond represents pooled effect size of long-term interventions; the width of diamond represents 95% confidence interval. The right side shows Hedges's g, 95% confidence interval, p-value for each category

Table 4 Meta-regression for non-linear (quadratic) model

Variable	Estimated Coefficient	SE	t	*p*
Intercept	−2.35	1.06	−2.21	0.05
Session time	0.03	0.05	0.59	0.57
Session time²	−0.00	0.00	−0.51	0.62
Frequency	2.15	0.93	2.33	0.04*
Frequency²	−0.57	0.23	−2.51	0.03*
Intervention Duration	0.09	0.05	1.70	0.12
Intervention Duration²	−0.00	0.00	−1.22	0.25

Significance code: *$p<0.05$; SE: residual standard error; model was weighted by inverse-variance

(SMD = 0.86, 95% CI [0.444, 1.277]) through trim-and-fill analysis was consistent with the original finding.

The leave-one-out (LOO) sensitivity analysis was performed to provide information about how individual studies might affect the overall effect size in meta-analysis. By removing one single study at a time, the result still remained consistent, supporting the reliability of the findings (see Fig. 6).

Discussion

This meta-analysis aimed to explore the effect of targeted PA interventions and the impact of intervention parameters on IC in children and adolescents with ADHD. The results showed that targeted PA interventions could improve IC in this population. Specifically, long-term interventions brought about a significant improvement in IC, but acute interventions did not. For intervention parameters, although subgroup meta-analyses of long-term interventions did not yield statistically significant between-group differences, examination of individual studies in each subgroup identified a moderate effect. Using meta-regression, we found an inverted U-shaped relationship between the intervention effect and PA frequency, with an estimated maximum effect occur at approximately 2 times/week. These findings support the effectiveness of targeted PA interventions and provide evidence for designing effective interventions.

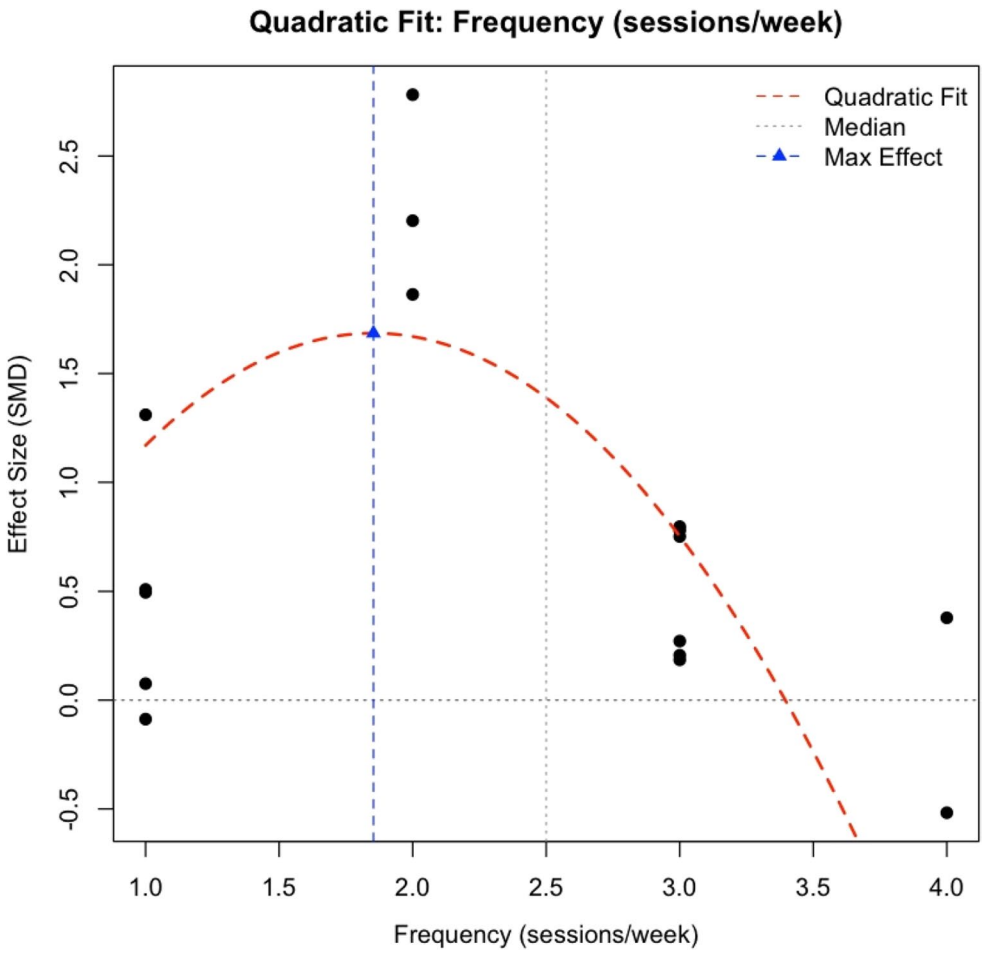


Fig. 4 Quadratic regression model results. The dashed curve line represents the quadratic fitted line, the blue triangle on the cross represents the predicted maximum effect, and the dashed vertical line shows the median frequency

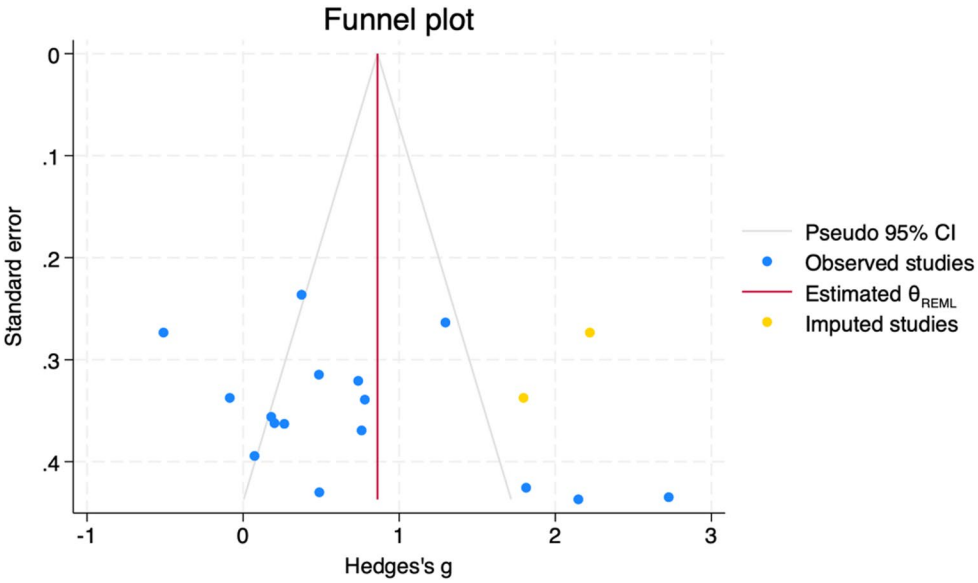


Fig. 5 The funnel plot for publication bias. The x-axis represents standardized hedges's g of each study; y-axis represents standard error; dots represent observed studies; squares are imputed studies estimated by Duval and Tweedie's trim-and-fill method; the vertical bar represents the weighted average of the overall effect; the funnel lines represent 95% confidence interval

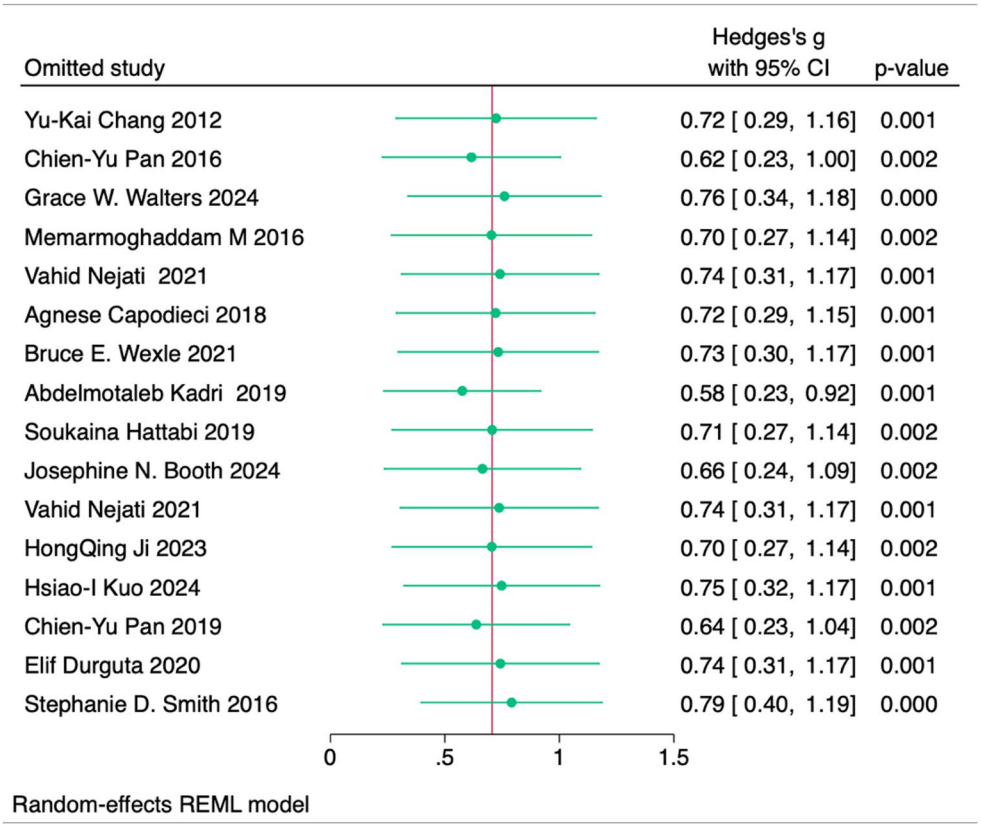


Fig. 6 Leave-one-out (loo) sensitivity analysis. Each row represents the pooled standardized effect size and 95% confidence interval after remove one study showed in the left side. The vertical line represents the overall summary effect

Overall effect of targeted physical activity intervention

Our meta-analysis revealed that the targeted PA interventions yielded a small-to-moderate positive effect on IC in children and adolescents with ADHD. This finding is consistent with previous systematic review and meta-analysis studies [14, 24, 58, 97–99]. The underlying mechanisms might involve an increased level of brain-derived neurotrophic factor (BDNF), which is a protein and plays a key role for neuronal growth, survival, and plasticity [100]. Individuals with ADHD may exhibit reduced BDNF

signaling, particularly in the prefrontal cortex associated with attention regulation and executive function [2, 101]. Regular participation in MVPA has been shown to upregulate BDNF expression, which may help enhance neural connectivity and cognitive flexibility [96]. Also, PA is known to increase the release of neurotransmitters such as dopamine and norepinephrine, both of which are essential in executive functioning and are dysregulated in individuals with ADHD [102]. Moreover, PA interventions could enhance cerebral blood flow that carries oxygen and nutrients to the brain, improving metabolic resources needed to support higher-level cognitive processing like IC in children and adolescents with ADHD [32].

Interpretations of heterogeneity

The overall meta-analysis revealed high heterogeneity (I^2 = 83.11%), suggesting significant variability in the intervention effects across studies. To better understand this variability, we firstly examined the acute and long-term intervention groups separately. The acute intervention group showed no heterogeneity (I^2 = 0%), indicating consistent effects across one-time interventions, and this result made further moderation analysis unnecessary. On the contrary, analysis of the long-term intervention group revealed high heterogeneity (I^2 = 84.96%), warranting further exploration of moderators. Subsequent subgroup meta-analyses based on session time, frequency, and intervention duration, however, did not yield statistically significant between-group difference (p >0.05), possibly due to the limited number of studies within subgroups. However, examination of individual studies in each subgroup resulted in statistical significance, indicating the intervention delivered 2–3 times per week, lasting around 60 min, and with intervention duration of ≥ 8 weeks were better than other categories. In addition, meta-regression analysis found an inverted-U relationship between frequency and intervention effect, suggesting that PA interventions at moderate frequencies may be more beneficial for cognitive outcomes, while those at very low or high frequencies might be less effective. Although this finding is in line with neurocognitive theories on arousal and cognitive load [68], it should be interpreted with caution due to the limited number of studies in each frequency category.

Analysis of intervention parameters

Intensity Due to the limited number of eligible studies, we conducted descriptive summary for intervention intensity in long-term interventions. The finding suggests that MVPA might be effective in enhancing IC in youth with ADHD. This is consistent with previous studies emphasizing the effect of higher intensity PA for overall executive function in ADHD [26]. From the perspective of neurocognitive mechanisms, MVPA can activate the function of the prefrontal cortex and stimulate the release of dopamine, norepinephrine, and BDNF to promote neuroplasticity [15, 16, 100, 101]. Excessive stimuli, on the other hand, may cause physical fatigue and deplete cognitive resources for IC [33, 103]. LPA (e.g., Yoga, Tai Chi) also showed a measurable improvement in IC [30, 31], and this effect may be explained by sensorimotor integration and cognitive processes. Both of them emphasize cognitive challenges rather than demanding high levels of intensity [33]. However, these findings should be interpreted with caution due to the nature of descriptive summary that limits the strength of causal inference.

Time parameters This meta-analysis explored the dose-response relationship between PA intervention parameters and IC in children and adolescents with ADHD. While group differences for session time, frequency, and intervention duration were not statistically significant, examination of individual studies in each subgroup allowed us to observe a moderate effect. Specifically, interventions with session time around 60 min, ≤ 3 times/week, and duration more than 8 weeks seemed to be more effective to improve IC than other categories. Meta-regression analyses showed that an inverted U-shaped relationship between PA frequency and intervention effect size.

For session time, individual studies with interventions around 60 min per session showed a large effect. This is aligned with the recommended exercise time per day by the WHO guidelines [35]. Likewise, Tomporowski and collogues found that around an hour of PA allowed children and adolescents with ADHD to raise the arousal level that was conducive to cognitive improvement without overstimulation [37]. From a neurocognitive viewpoint, moderate session time (e.g., around 60 min) is sufficient to bring nerve stimulation (e.g., increased cerebral blood flow) while avoiding attention exhaustion [32]. In this study, shorter sessions seemed unable to provide sufficient stimulus for the improvement of cognitive functioning, and longer sessions led to fatigue that could diminish the effectiveness of PA [27, 33]. Meanwhile, based on the dual-task load theory, since executive function is an advanced cognitive function, the effect of PA will be affected by the allocation of limited cognitive and attentional resources [103]. If the cognitive load of the task (e.g., higher intensity activity, complex motor skill, or longer time per session) is too high, it may lead to the depletion of cognitive

resources and then hinder the improvement of executive functioning [104].

For frequency, meta-regression results revealed an inverted U-shaped relationship, with optimal intervention effects observed around 2 times per week. This finding indicates that the frequency of 1–3 sessions/week was most effective for the improvement of IC, which aligns with neurocognitive models suggesting that moderate frequencies may yield cognitive benefits by balancing neural stimulation with adequate recovery [2, 17]. Although daily PA may theoretically improve executive function, practical factors, such as fatigue and adherence, may have a negative impact on the effectiveness of PA [105]. With 1–3 session per week, the intervention could provide sufficient time for children and adolescents with ADHD to recover from mental and physical fatigue and maintain engagement over time. A higher frequency, however, may lead to overstimulation and inadequate recovery and therefore diminish the intervention's cognitive benefits [12, 27, 33]. The U-shaped relationship between frequency and intervention effects, however, should be interpreted with caution. Because the number of studies within each frequency category was limited and unequal, the statistical power to detect robust moderator effects was constrained. This relationship may be also influenced by unmeasured confounding factors. For instance, participant characteristics (e.g., age, gender, ADHD severity, or baseline executive function) are known to moderate intervention effects but were not consistently reported across included studies [106]. In addition, variation in the types of PA may require different optimal frequencies to achieve cognitive benefits [14, 38]. Due to limited reporting in included studies, we were unable to examine the interaction between frequency and these potential moderators in the present study. Future research might use muti-arm randomized designs to explore these potential interactive relationships.

Regarding intervention duration, while acute interventions appeared to have no significant impacts on IC in children and adolescents with ADHD, long-term interventions brought about a moderate to large improvement. While some studies claimed that acute interventions (e.g., acute aerobic exercise with 30 min) could temporarily improve IC [17, 41, 107], this meta-analysis did not find a significant effect, which could be explained by the temporarily increase in neurotransmitter levels [108]. Since children and adolescents with ADHD metabolize neurotransmitters more quickly, a single PA may be insufficient to bring the cognitive improvement [109]. Moreover, IC is complex executive function involving sustained attention, cognitive flexibility, and inhibition, requiring repeated activation and strengthening of relevant neural pathways to observe change [2]. The prefrontal cortex, critical for IC, undergoes slow and experience-dependent plasticity, which is less likely to be shaped by short-term interventions [23]. In contrast, long-term interventions provide sufficient time and repeated practice for children and adolescents with ADHD, contributing to neural adaptations and consolidate the improvement in IC [40]. Through repeated and sustained PA interventions, the increase in the activation level of the prefrontal cortex in children with ADHD directly contributes to IC improvement and neuroplasticity [16, 58, 100]. Another explanation may lie in the limited number of eligible acute intervention studies (k = 3) that has restricted statistical power. Moreover, all included acute interventions had a relatively small sample size, inflating the risk of imprecise estimates and wide confidence intervals. Therefore, while acute interventions may hold some promise in empirical studies, current meta-analytic evidence is insufficient to draw a firm conclusion. Future research with large sample can continue to explore the effect of different intervention design in children and adolescents with ADHD.

The findings of this meta-analysis are consistent with those of previous studies, demonstrating that PA interventions could improve executive function, particularly IC, in children and adolescents with ADHD [24, 58, 99, 106]. Several meta-analysis studies found that targeted PA interventions, such as cognitively engaging PA or structured programs, were associated with improvement of executive function [24, 38, 110]. Compared to these reviews, the present study focused on IC as a specific executive function and investigated how intervention parameters, particularly for session time, frequency, intervention duration, may moderate intervention effectiveness. While most previous reviews examined general effect of PA or compared types of PA, few studies explored how specific intervention parameters influence outcomes. Specifically, researchers highlighted the potential of MVPA but did not quantify the dose-response relationship between intervention parameters and cognitive outcomes [26, 111]. Our study adds new findings by identifying an inverted U-shaped relationship between frequency and intervention effect, suggesting that PA with moderate frequencies may be more beneficial than that with higher

or lower frequencies. In addition, examination of individual studies in subgroup meta-analyses suggested that interventions with session time around 60 min and intervention duration of at least 8 weeks may bring cognitive benefits, which offers preliminary evidence and indicates potential directions for future research.

Limitations and implications

While providing valuable insights into how PA interventions and parameters affect IC in children and adolescents with ADHD, this study has certain limitations. First, a relatively large number of full-text articles were inaccessible and therefore excluded, which may have introduced selection bias and limited the comprehensiveness of our synthesis. Second, the small number of studies included in subgroup meta-analyses, particularly those reporting intensity, limited the statistical power to detect moderator effects. Also, due to the limited and unequal numbers within each category, we were unable to conduct subgroup meta-analysis or meta-regression analysis on PA intensity. Future research may include more studies with reported intervention parameters to further investigate dose-response relationships. Detailed reporting of exercise characteristics, such as objective intensity measures (e.g., heart rate, RPE), would allow for more accurate dose-response estimations. Future studies are also recommended to adopt multi-arm randomized controlled trials comparing specific intervention parameters (e.g., different frequency, session time, or intensity level. Third, as stated in the introduction, we did not include PA types as a moderator in this study, because previous meta-analysis studies have already examined but did not find any moderation effect of PA on IC [14]. Also, the total number of empirical studies included in the present meta-analysis is relatively small. The types of PA employed in these studies are even few and far between. Therefore, dividing these studies based on PA types would make the number in the subcategories unbalanced and negligible. Consequently, continuing with subgroup- or meta-regression analyses would compromise statistical power and may yield misleading results. Nevertheless, we recognize the value of this variable and encourage future studies to include PA types for a more inclusive review once a sufficient number of empirical studies are available. Fourth, this study only focused on four intervention parameters but did not explore other moderating factors such as age or gender. This is because most studies did not report data for different age or gender groups separately. Therefore, we were unable to extract sufficient data for these variables. Meanwhile, small number of available studies further limited the feasibility of subgroup analyses. Future research can examine how these variables interact with each other. Finally, this study did not account for different ADHD subtypes or severity levels, again due to the limited number of studies reporting ADHD subtypes, which may have moderated the effectiveness of interventions. Further studies may take the ADHD subtype or severity into consideration to establish effective PA intervention protocols. Nevertheless, this meta-analysis provides new insights into and evidence for intervention design to improve IC in children and adolescents with ADHD. These studies would facilitate the development of more precise and personalized PA intervention protocols for youth with ADHD, and assist practitioners to design intervention therapies for and enhance executive function in children and adolescents with ADHD.

Conclusion

This meta-analysis found targeted PA interventions were associated with improvements in IC in children and adolescents with ADHD. Long-term interventions appeared to be more beneficial than acute interventions. Although group differences for session time, frequency and intervention duration in long-term interventions did not yield statistical significance, examination of individual empirical studies suggested that moderate frequencies, session time around 60 min, and longer intervention duration may be associated with more favorable outcomes. Additionally, meta-regression indicated a potential inverted U-shape relationship between frequency and intervention effect, which may need further investigation. Finally, we hope this meta-analysis may provide preliminary insight to motivate future researchers invest more effort in this area. Meanwhile, the results may assist practitioners in designing PA-based intervention programs for children and adolescents with ADHD.

Appendix

Table 5 Full search term and search strategy

	Key Word	MeSH	Full Search Term
#1	Physical activity	Exercise	(Exercises OR Exercise Physical OR Exercises Physical OR Physical Exercise OR Physical Exercises OR Physical Activity OR Activities, Physical OR Activity, Physical OR Physical Activities OR Exercise, Aerobic OR Aerobic Exercise OR Aerobic Exercises OR Exercises, Aerobic OR Exercise, Isometric OR Exercises, Isometric OR Isometric Exercises OR Isometric Exercise OR Acute Exercise OR Acute Exercises OR Exercise, Acute OR Exercises, Acute OR Exercise Training OR Exercise Trainings OR Training, Exercise OR Trainings, Exercise) AND
#2	Executive function	Executive Function	(Executive Functions OR Function, Executive OR Functions, Executive OR Executive Control OR Executive Controls) AND
#3	Attention deficit hyperactivity disorder	Attention deficit hyperactivity disorder	(ADHD OR ADDH OR Attention Deficit Disorders with Hyperactivity OR Attention Deficit Hyperactivity Disorders OR Attention Deficit Hyperactivity Disorder OR Attention Deficit-Hyperactivity Disorder OR Attention Deficit-Hyperactivity Disorders OR Deficit-Hyperactivity Disorder, Attention OR Deficit-Hyperactivity Disorders, Attention OR Disorder, Attention Deficit-Hyperactivity OR Disorders, Attention OR Deficit-Hyperactivity OR Hyperkinetic Syndrome OR Syndromes, Hyperkinetic OR Attention Deficit Disorder OR Attention Deficit Disorders OR Deficit Disorder, Attention OR Deficit Disorders, Attention OR Disorder, Attention Deficit OR Disorders, Attention Deficit OR Brain Dysfunction, Minimal OR Dysfunction, Minimal Brain OR Minimal Brain Dysfunction) AND
#4	Children	Child	(child OR children) AND
#5	Adolescents	Adolescents	(adolescents or teenagers or young adults or teen or youth or student or adolescence)
Final search strategy: #1+#2+#3+#4+#5			

MeSH Medical term. The search strategy used the Boolean Operator (AND, OR) for keywords combination. There is not time and Language limitation

Table 6 The characteristics of included studies

Study	Region	*N* (Male)	Age (years)	Intervention (*N*)	Control (*N*)	Intensity	Time (min)	F(times)	Duration (week)
Yu-Kai Chang 2012 [17]	Taiwan, China	40(37)	10.43 ± 0.9	Aerobic run on treadmill (*N*=20)	Watching running video (*N*=20)	MPA	30	1	1
Chien-Yu Pan 2016 [86]	Taiwan, China	32(32)	9.08 ± 1.43	Racket-sport intervention (*N*=16)	Daily routine (*N*=16)	N/A	70	2	12
Grace W. Walters 2024 [41]	UK	34(30)	9.9 ± 1.0	Circuit-based cognitively engaging exercise (*N*=19)	Waitlist control (*N*=15)	N/A	30	1	1
Memarmoghaddam M 2016 [39]	Iran	36(36)	8.31 ± 1.29	Cognitive engaging program (*N*=19)	Daily routine (*N*=17)	MVPA	90	3	8

N sample size; *HRmax* maximum heart rate; *mins* minutes; *F* frequency; *LPA* low intensity; *MPA* moderate intensity physical activity; *MVPA* moderate-to-vigorous intensity physical activity; *N/A* study did not report and evaluate intervention intensity

Table 7 The characteristics of included studies (continued)

Study	Region	*N* (Male)	Age (years)	Intervention (*N*)	Control (*N*)	Intensity (HRmax)	Time (min)	F(times)	Duration (week)
Vahid Nejati 2021 [88]	Iran	26(26)	9.43 ± 1.43	Exercise for Cognitive Improvement and Rehabilitation (*N* = 15)	Simple aerobic running (*N* = 14)	N/A	50	3	4
Agnese Capodiecl 2019 [89]	Italy	12(9)	7.23 ± 0.62	Cognitive Training Activities (*N* = 12)	Waitlist control (*N* = 12)	N/A	60	1	8
Bruce E. Wexle [90] 2021	USA	73(48)	7.4 ± 1.2	Computer-presented cognitive-training exercises (cognitive training) (*N* = 31)	Daily routine (*N* = 42)	N/A	45	4	15
Abdelmotaleb Kadri 2019 [91]	Tunisia	40(36)	14.5 ± 3.5	Taekwondo (TKD) intervention (*N* = 20)	Daily routine (*N* = 20)	N/A	50	2	72

N sample size; *HRmax* maximum heart rate; *mins* minutes; *F* frequency; *LPA* low intensity; *MPA* moderate intensity physical activity; *MVPA* moderate-to-vigorous intensity physical activity; *N/A* study did not report and evaluate intervention intensity

Table 8 The characteristics of included studies (continued)

Study	Region	*N* (Male)	Age (years)	Intervention (*N*)	Control (*N*)	Intensity (HRmax)	Time (min)	F(times)	Duration (week)
Soukaina Hattabi 2019 [40]	Tunisia	40(35)	9.95 ± 1.31	Water exercise program (*N* = 20)	Daily routine (*N* = 20)	MVPA	90	3	12
Josephine N. Booth 2024 [92]	UK	15(10)	10.75 ± 0.83	Cognitive game program (*N* = 31)	Daily routine (*N* = 38)	MVPA	60	1	12
Vahid Nejati 2021 [88]	Iran	35(16)	9.73 ± 1.94	Balance-based Attentive Rehabilitation of Attention Networks (*N* = 15)	Running (*N* = 14)	N/A	50	3	5
HongQing Ji 2023 [59]	Korea	30(26)	9 ± 1.46	Exergaming (*N* = 16)	Bicycle exercise (*N* = 14)	MVPA	50	3	4

N sample size; *HRmax* maximum heart rate; *mins* minutes; *F* frequency; *LPA* low intensity; *MPA* moderate intensity physical activity; *MVPA* moderate-to-vigorous intensity physical activity; *N/A* study did not report and evaluate intervention intensity

Table 9 The characteristics of included studies (continued)

Study	Region	*N* (Male)	Age (years)	Intervention (*N*)	Control (*N*)	Intensity (HRmax)	Time (min)	F(times)	Duration (week)
Hsiao-I Kuo 2024 [94]	Taiwan, China	24(13)	17.2 ± 2.2	Aerobic exercise intervention-stationary exercise bike (*N* = 12)	Waitlist control (*N* = 12)	MPA	30	1	1
Chien-Yu Pan 2019 [87]	Taiwan, China	30(30)	8.93 ± 1.49	Table tennis program merging physical and cognitive training (*N* = 15)	Daily routine (*N* = 15)	N/A	70	2	12
Elif Durguta 2020 [95]	Turkey	30(24)	8.33 ± 1.17	Whole body vibration with treadmill training group (*N* = 15)	Treadmill training (*N* = 15)	N/A	45	3	8
Stephanie D. Smith 2016	USA& China	112(78)	7.4 ± 1.1	Integrated Brain, Body, and Social (IBBS) Intervention (*N* = 48)	Control (as usual) (*N* = 44)	MPA	120	4	15

N sample size; *HRmax* maximum heart rate; *mins* minutes; *F* frequency; *LPA* low intensity; *MPA* moderate intensity physical activity; *MVPA* moderate-to-vigorous intensity physical activity; *N/A* study did not report and evaluate intervention intensity

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Author contributions Yiwei Min designed this study, conducted data analysis and composes the manuscript. Yuxin Chen screened articles and contributed to the conceptualization of the study. As the senior author, Dr. Jiling Liu was responsible for directing the study, guiding the manuscript writing process, reviewing and editing the final manuscript.

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Data availability The authors confirm that all data analyzed during this study are included in this published article and all secondary sources and data supporting the findings of this study were all publicly available at the time of submission.

Declarations

Competing interests The authors declare no competing interests.

Conflict of interest All authors declare no conflicts of interest relevant to this research.

References

1. Barkley RA, Clarkin JF, Kernberg OF (n.d.) Attention-deficit hyperactivity disorder: a handbook for diagnosis and treatment, 3rd ed
2. Diamond A (2013) Executive functions. Annu Rev Psychol 64:135–168
3. Best JR, Miller PH (2010) A developmental perspective on executive function. Child Dev 81(6):1641–1660
4. Ordaz SJ, Foran W, Velanova K, Luna B (2013) Longitudinal growth curves of brain function underlying inhibitory control through adolescence. J Neurosci 33:18109–18124
5. Interactions between the emotional and executive brain systems [Internet] (2023). Wikipedia [cited 2025 Aug 27]. Available from: https://en.wikipedia.org/w/index.php?title=Interactions_between_the_emotional_and_executive_brain_systems&oldid=1141415820
6. Liew J (2012) Effortful control, executive functions, and education: bringing self-regulatory and social-emotional competencies to the table. Child Dev Perspect 6:105–111
7. Privitera AJ, Zhou Y, Xie X (2023) Inhibitory control as a significant predictor of academic performance in Chinese high schoolers. Child Neuropsychol 29:457–473
8. Thapar A, Cooper M (2016) Attention deficit hyperactivity disorder. Lancet 387:1240–1250
9. Shaw M, Hodgkins P, Caci H, Young S, Kahle J, Woods AG et al (2012) A systematic review and analysis of long-term outcomes in attention deficit hyperactivity disorder: effects of treatment and non-treatment. BMC Med 10:99
10. Pelham WE Jr., Fabiano GA (2008) Evidence-based psychosocial treatments for attention-deficit/hyperactivity disorder. J Clin Child Adolesc Psychol 37:184–214
11. Cortese S, Adamo N, Giovane CD, Mohr-Jensen C, Hayes AJ, Carucci S et al (2018) Comparative efficacy and tolerability of medications for attention-deficit hyperactivity disorder in children, adolescents, and adults: a systematic review and network meta-analysis. Lancet Psychiatry 5:727–738
12. Ludyga S, Gerber M, Mucke M, Brand S, Weber P, Brotzmann M et al (2020) The acute effects of aerobic exercise on cognitive flexibility and task-related heart rate variability in children with ADHD and healthy controls. J Atten Disord 24:693–703
13. Elmaghraby R, Pines A, Geske JR, Coombes BJ, Leung JG, Croarkin PE et al (2024) Risk of newly diagnosed psychotic symptoms in youth receiving medications for attention-deficit/hyperactivity disorder. Jaacap Open 2:135–144
14. Song Y, Fan B, Wang C, Yu H (2023) Meta-analysis of the effects of physical activity on executive function in children and adolescents with attention deficit hyperactivity disorder. PLoS ONE 18:e0289732
15. Ludyga S, Gerber M, Brand S, Holsboer-Trachsler E, Pühse U (2016) Acute effects of moderate aerobic exercise on specific aspects of executive function in different age and fitness groups: a meta-analysis. Psychophysiology 53:1611–1626
16. Currie J, Ramsbottom R, Ludlow H, Nevill A, Gilder M (2009) Cardio-respiratory fitness, habitual physical activity and serum brain derived neurotrophic factor (BDNF) in men and women. Neurosci Lett 451:152–155
17. Chang Y-K, Liu S, Yu H-H, Lee Y-H (2012) Effect of acute exercise on executive function in children with attention deficit hyperactivity disorder. Arch Clin Neuropsychol 27:225–237
18. Jun C (2023) Impacts of soccer on executive function in boys with adhd. Rev Bras Med Esporte [Internet]. Available from: https://www.cochranelibrary.com/central/doi/10.1002/central/CN-02561216/full
19. Qiu C, Zhai Q, Chen S (2024) Effects of practicing closed- vs. open-skill exercises on executive functions in individuals with attention deficit hyperactivity disorder (ADHD)—a meta-analysis and systematic review. Behav Sci 14:499
20. Zhu F, Zhu X, Bi X, Kuang D, Liu B, Zhou J et al (2023) Comparative effectiveness of various physical exercise interventions on executive functions and related symptoms in children and adolescents with attention deficit hyperactivity disorder: a systematic review and network meta-analysis. Front Public Health 11:1133727
21. Liang X, Li R, Wong SHS, Sum RKW, Sit CHP (2021) The impact of exercise interventions concerning executive functions of children and adolescents with attention-deficit/hyperactive disorder: a systematic review and meta-analysis. Int J Behav Nutr Phys Act 18:68
22. Physical activity and your heart - types | NHLBI, NIH [Internet] (2022) [cited 2025 June 24]. Available from: https://www.nhlbi.nih.gov/health/heart/physical-activity/types
23. Best JR (2010) Effects of physical activity on children's executive function: contributions of experimental research on aerobic exercise. Dev Rev 30:331–351
24. Sun F, Fang Y, Chan CKM, Poon ETC, Chung LMY, Or PPL et al (2024) Structured physical exercise interventions and children and adolescents with attention deficit hyperactivity disorder: a systematic review and meta-analysis. Child Care Health Dev 50:e13150
25. Benzing V, Chang Y-K, Schmidt M (2018) Acute physical activity enhances executive functions in children with ADHD. Sci Rep 8:12382
26. Chen J-W, Du W-Q, Zhu K (2024) Optimal exercise intensity for improving executive function in patients with attention deficit hyperactivity disorder: systematic review and network meta-analysis. Eur Child Adolesc Psychiatry. https://doi.org/10.1007/s00787-024-02507-6
27. Mehren A, Diaz Luque C, Brandes M, Lam AP, Thiel CM, Philipsen A et al (2019) Intensity-dependent effects of acute exercise on executive function. Neural Plast 2019:8608317
28. Tsai Y-J, Hsieh S-S, Huang C-J, Hung T-M (2021) Dose-response effects of acute aerobic exercise intensity on inhibitory control in

children with attention deficit/hyperactivity disorder. Front Hum Neurosci 15:617596
29. Tempest GD, Eston RG, Parfitt G (2014) Prefrontal cortex haemodynamics and affective responses during exercise: a multichannel near infrared spectroscopy study. PLoS ONE 9:e95924
30. Azami S, Alimadadi Z, Ahmadi A, Hemmati F, Mirmohammad M, Mashayekhi R (2023) The efficacy of cognitive-motor rehabilitation on cognitive functions and behavioral symptoms of attention deficit/hyperactivity disorder (ADHD) children: Specification of near-transfer and far-transfer effects in comparison to medication. J Educ Health Promot 12(1):64
31. Chou C-C, Huang C-J (2017) Effects of an 8-week yoga program on sustained attention and discrimination function in children with attention deficit hyperactivity disorder. PeerJ 5:e2883
32. Hillman CH, Erickson KI, Kramer AF (2008) Be smart, exercise your heart: exercise effects on brain and cognition. Nat Rev Neurosci 9:58–65
33. Wymbs F, Wymbs B, Margherio S, Burd K (2021) The effects of high intensity versus low intensity exercise on academic productivity, mood, and behavior among youth with and without ADHD. J Child Fam Stud 30:460–473
34. Vasilopoulos F, Jeffrey H, Wu Y, Dumontheil I (2023) Multi-level meta-analysis of physical activity interventions during childhood: effects of physical activity on cognition and academic achievement. Educ Psychol Rev 35:59
35. WHO guidelines on physical activity and sedentary behaviour (2020) 1st ed. Geneva: World Health Organization
36. Donnelly JE, Hillman CH, Castelli D, Etnier JL, Lee S, Tomporowski P et al (2016) Physical activity, fitness, cognitive function, and academic achievement in children: a systematic review. Med Sci Sports Exerc 48:1197–1222
37. Verret C, Guay M-C, Berthiaume C, Gardiner P, Béliveau L (2012) A physical activity program improves behavior and cognitive functions in children with ADHD: an exploratory study. J Atten Disord 16:71–80
38. Tan B, Pooley J, Speelman C (2016) A meta-analytic review of the efficacy of physical exercise interventions on cognition in individuals with autism spectrum disorder and ADHD. J Autism Dev Disord 46:3126–3143
39. Memarmoghaddam M, Torbati H, Sohrabi M, Mashhadi A, Kashi A (2016) Effects of a selected exercise programon executive function of children with attention deficit hyperactivity disorder. J Med Life 9:373–379
40. Hattabi S, Bouallegue M, Ben Yahya H, Bouden A (2019) Rehabilitation of ADHD children by sport intervention: a Tunisian experience. Tunis Med 97:874–881
41. Walters GW, Taylor S, Sweeney EL, Cooper SB, Williams RA, Dring KJ (2024) Effect of an acute bout of exercise on executive function and sleep in children with attention deficit hyperactivity disorder and autism spectrum disorder. Ment Health Phys Act 27:100624
42. Chang Y-K, Hung C-L, Huang C-J, Hatfield BD, Hung T-M (2014) Effects of an aquatic exercise program on inhibitory control in children with ADHD: a preliminary study. Arch Clin Neuropsychol 29:217–223
43. Tomporowski PD, McCullick B, Pendleton DM, Pesce C (2015) Exercise and children's cognition: the role of exercise characteristics and a place for metacognition. J Sport Health Sci 4:47–55
44. Bull FC, Al-Ansari SS, Biddle S, Borodulin K, Buman MP, Cardon G et al (2020) World health organization 2020 guidelines on physical activity and sedentary behaviour. [cited 2025 Mar 17]. Available from: https://bjsm.bmj.com/content/54/24/1451?fbclid=IwAR1Xxfw33ABqf6iZJDm8bWVRUthXGR8bquQVZ2z6WmhAWl4nm6v_TZUrMGU
45. Page MJ, Moher D, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD et al (2021) PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. BMJ 372:n160
46. Moseley AM, Herbert RD, Sherrington C, Maher CG (2002) Evidence for physiotherapy practice: a survey of the physiotherapy evidence database (PEDro). Aust J Physiother 48:43–49
47. Maher CG, Sherrington C, Herbert RD, Moseley AM, Elkins M (2003) Reliability of the PEDro scale for rating quality of randomized controlled trials. Phys Ther 83:713–721
48. Goulas G (2022) How can I cite covidence? [Internet]. [cited 2025 June 17]. Available from: https://support.covidence.org/help/how-can-i-cite-covidence
49. Wan X, Wang W, Liu J, Tong T (2014) Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. BMC Med Res Methodol 14:135
50. Higgins JPT, Cochrane C (eds) (2019) Cochrane handbook for systematic reviews of interventions. Second edition. Hoboken, NJ: Wiley-Blackwell
51. Norton K, Norton L, Sadgrove D (2010) Position statement on physical activity and exercise intensity terminology. J Sci Med Sport 13:496–502
52. Medicine AC (2017) Of S ACSM's exercise testing and prescription. Lippincott Williams & Wilkins
53. Zhu L, Li L, Wang L, Jin X, Zhang H (2020) Physical activity for executive function and activities of daily living in AD patients: a systematic review and meta-analysis. Front Psychol [Internet]. [cited 2025 Mar 22];11. Available from: https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2020.560461/full
54. Tian S, Liang Z, Qiu F, Wang X (2023) Physical activity on executive function in sedentary individuals: systematic review and meta-analysis of randomized controlled trials. PLoS ONE 18:e0294251
55. Rathore A, Lom B (2017) The effects of chronic and acute physical activity on working memory performance in healthy participants: a systematic review with meta-analysis of randomized controlled trials. Syst Rev 6:124
56. Muntaner-Mas A, Morales JS, Martínez-de-Quel Ó, Lubans DR, García-Hermoso A (2024) Acute effect of physical activity on academic outcomes in school-aged youth: a systematic review and multivariate meta-analysis. Scand J Med Sci Sports 34:e14479
57. Li JW, O'Connor H, O'Dwyer N, Orr R (2017) The effect of acute and chronic exercise on cognitive function and academic performance in adolescents: a systematic review. J Sci Med Sport 20:841–848
58. Liu S, Yu Q, Li Z, Cunha PM, Zhang Y, Kong Z et al (2020) Effects of acute and chronic exercises on executive function in children and adolescents: a systemic review and meta-analysis. Front Psychol [Internet]. [cited 2025 Mar 22];11. Available from: https://www.frontiersin.org/journals/psychology/articles/https://doi.org/10.3389/fpsyg.2020.554915/full
59. Ji H, Wu S, Won J, Weng S, Lee S, Seo S et al (2023) The effects of exergaming on attention in children with attention deficit/hyperactivity disorder: randomized controlled trial. JMIR Serious Games 11:e40438
60. Bhogal SK, Teasell RW, Foley NC, Speechley MR (2005) The PEDro scale provides a more comprehensive measure of methodological quality than the Jadad scale in stroke rehabilitation literature. J Clin Epidemiol 58:668–673
61. Hedges LV, Olkin I (1985) Chapter 5 - estimation of a single effect size: parametric and nonparametric methods. In: Hedges LV, Olkin I, editors. Stat Methods Meta-Anal [Internet]. San Diego: Academic Press; [cited 2025 June 18]. pp. 75–106. Available from: https://www.sciencedirect.com/science/article/pii/B9780080570655500105
62. Biondi Zoccai G (2014) Network meta-analysis: evidence synthesis with mixed treatment comparison [Internet]. USA; [cited 2024

Nov 3]. Available from: https://iris.uniroma1.it/handle/11573/606605
63. Hedges LV, Olkin I (1985) Chapter 9 - random effects models for effect sizes. In: Hedges LV, Olkin I, editors. Stat Methods Meta-Anal [Internet]. San Diego: Academic Press; [cited 2025 June 18]. pp. 189–203. Available from: https://www.sciencedirect.com/science/article/pii/B9780080570655500142
64. DerSimonian R, Laird N (1986) Meta-analysis in clinical trials. Control Clin Trials 7:177–188
65. Higgins JPT, Thompson SG (2002) Quantifying heterogeneity in a meta-analysis. Stat Med 21:1539–1558
66. Hill M (2024) Cochrane handbook for systematic reviews of interventions
67. Diamond A (2015) Effects of physical exercise on executive functions: going beyond simply moving to moving with thought. Ann Sports Med Res 2:1011
68. Chang Y-K, Chu C-H, Wang C-C, Wang Y-C, Song T-F, Tsai C-L et al (2015) Dose-response relation between exercise duration and cognition. Med Sci Sports Exerc 47:159–165
69. Duval S, Tweedie R (2000) Trim and fill: a simple Funnel-plot–based method of testing and adjusting for publication bias in meta-analysis. Biometrics 56:455–463
70. Patsopoulos NA, Evangelou E, Ioannidis JP (2008) Sensitivity of between-study heterogeneity in meta-analysis: proposed metrics and empirical evaluation. Int J Epidemiol 37:1148–1157
71. Nair AS, Borkar N (2024) Sensitivity and subgroup analysis in a meta-analysis – what we should know? Indian J Anaesth 68:922–924
72. Bustamante EE, Santiago-Rodriguez ME, Ramer JD, Balbim GM, Mehta TG, Frazier SL (2019) Physical activity and ADHD: evidence on development, transient and durable neurocognitive effects, and applications. Pensar En Mov Rev Cienc Ejerc Salud 17:e34662–e34662
73. Liang X, Sit CHP, Li R, Qiu H, Wang P, Yang B et al (2022) The effect of aerobic-and neurocognitive exercise on executive functions and sleep in children with ADHD: 949. Med Sci Sports Exerc 54:233
74. Flynn RM, Colon N (2016) Solitary active videogame play improves executive functioning more than collaborative play for children with special needs. Games Health J 5:398–404
75. Mazzoli E, Salmon J, Teo W-P, Pesce C, He J, Ben-Soussan TD et al (2021) Breaking up classroom sitting time with cognitively engaging physical activity: behavioural and brain responses. PLoS ONE 16:e0253733
76. Tseng Y-H, Chao H-H, Hung C-L (2022) Effect of a strategic physical activity program on cognitive flexibility among children with internet addiction: a pilot study. Children (Basel) 9:798
77. Ji C, Yang J, Lin L, Chen S (2022) Executive function improvement for children with autism spectrum disorder: a comparative study between virtual training and physical exercise methods. Children (Basel) 9:507
78. Clark D, Seymour KE, Findling RL, Mostofsky SH (2020) Subtle motor signs as a biomarker for mindful movement intervention in children with attention-deficit/hyperactivity disorder. J Dev Behav Pediatr 41:349–358
79. Jiang K, Xu Y, Li Y, Li L, Yang M, Xue P (2022) How aerobic exercise improves executive function in ADHD children: a resting-state fMRI study. Int J Dev Neurosci 82:295–302
80. Romero-Ayuso D, del Pino-González A, Torres-Jiménez A, Juan-González J, Celdrán FJ, Franchella MC et al (2024) Enhancing ecological validity: virtual reality assessment of executive functioning in children and adolescents with ADHD. Children (Basel) 11:986
81. Shema-Shiratzky S, Brozgol M, Cornejo-Thumm P, Geva-Dayan K, Rotstein M, Leitner Y et al (2019) Virtual reality training to enhance behavior and cognitive function among children with attention-deficit/hyperactivity disorder: brief report. Dev Neurorehabil 22:431–436
82. Vibholm HA, Pedersen J, Holm A, Krue S (2014) A multicenter, randomized, controlled study of training executive, attention, and motor skills (TEAMS) in Danish preschool children with attention-deficit/hyperactivity disorder: rationale and description of the intervention and study protocol. Scand J Child Adolesc Psychiatr Psychol 2:2–10
83. Benzing V, Schmidt M (2017) Cognitively and physically demanding exergaming to improve executive functions of children with attention deficit hyperactivity disorder: a randomised clinical trial. BMC Pediatr 17:8
84. de la Merced García C, Planillo Jarauta E (2022) Efectos de Los bici-pupitres sobre estudiantes Con Trastorno Por déficit de atención e hiperactividad. (spanish). Rev Investig En Educ 20:223–239
85. Imola B (2023) A mindfulness-alapú fejlesztés Folyamata és hatásmechanizmusa Az ADHD-val rendelkező tanulókra. Iskolakultúra 33:34–52
86. Pan C-Y, Chu C-H, Tsai C-L, Lo S-Y, Cheng Y-W, Liu Y-J (2016) A racket-sport intervention improves behavioral and cognitive performance in children with attention-deficit/hyperactivity disorder. Res Dev Disabil 57:1–10
87. Pan C-Y, Tsai C-L, Chu C-H, Sung M-C, Huang C-Y, Ma W-Y (2019) Effects of physical exercise intervention on motor skills and executive functions in children with ADHD: a pilot study. J Atten Disord 23(4):384–397
88. Nejati V (2021) Balance-based attentive rehabilitation of attention networks (BARAN) improves executive functions and ameliorates behavioral symptoms in children with ADHD. Complement Ther Med 60:102759
89. Capodieci A, Re AM, Fracca A, Borella E, Carretti B (2019) The efficacy of a training that combines activities on working memory and metacognition: transfer and maintenance effects in children with ADHD and typical development. J Clin Exp Neuropsychol 41(10):1074–1087
90. Wexler BE, Vitulano LA, Moore C, Katsovich L, Smith SD, Rush C et al (2021) An integrated program of computer-presented and physical cognitive training exercises for children with attention-deficit/hyperactivity disorder. Psychol Med 51:1524–1535
91. Kadri A, Slimani M, Bragazzi N, Tod D, Azaiez F (2019) Effect of taekwondo practice on cognitive function in adolescents with attention deficit hyperactivity disorder. Int J Environ Res Public Health 16:204
92. Booth JN, Mitchell IA, Tomporowski PD, McCullick BA, Boyle JME, Reilly JJ (2024) Evaluation of a pilot physical activity intervention for children with ADHD symptoms and reading difficulties. J Res Spec Educ Needs 24:254–268
93. Nejati V, Derakhshan Z (2021) The effect of physical activity with and without cognitive demand on the improvement of executive functions and behavioral symptoms in children with ADHD. Expert Rev Neurother 21:607–614
94. Kuo H-I, Sun J-L, Nitsche M, Chang J-C (2024) An investigation of the acute effects of aerobic exercise on executive function and cortical excitability in adolescents with attention deficit hyperactivity disorder (ADHD). Eur Child Adolesc Psychiatry. https://doi.org/10.1007/s00787-024-02467-x
95. Durgut E, Orengul AC, Algun ZC (2020) Comparison of the effects of treadmill and vibration training in children with attention deficit hyperactivity disorder: a randomized controlled trial. NeuroRehabilitation 47:121–131
96. Smith SD, Vitulano LA, Katsovich L, Li S, Moore C, Li F et al (2020) A randomized controlled trial of an integrated brain, body, and social intervention for children with ADHD. J Atten Disord 24:780–794

97. Welsch L, Alliott O, Kelly P, Fawkner S, Booth J, Niven A (2021) The effect of physical activity interventions on executive functions in children with ADHD: a systematic review and meta-analysis. Ment Health Phys Act 20:100379
98. Salari N, Ghasemi H, Abdoli N, Rahmani A, Shiri MH, Hashemian AH et al (2023) The global prevalence of ADHD in children and adolescents: a systematic review and meta-analysis. Ital J Pediatr 49:48
99. Xie Y, Gao X, Song Y, Zhu X, Chen M, Yang L et al (2021) Effectiveness of physical activity intervention on ADHD symptoms: a systematic review and meta-analysis. Front Psychiatry 12:706625
100. Huang T, Larsen KT, Ried-Larsen M, Møller NC, Andersen L (2014b) The effects of physical activity and exercise on brain-derived neurotrophic factor in healthy humans: a review. Scand J Med Sci Sports 24:1–10
101. Liu D-Y, Shen X-M, Yuan F-F, Guo O-Y, Zhong Y, Chen J-G et al (2015) The physiology of BDNF and its relationship with ADHD. Mol Neurobiol 52:1467–1476
102. Ellemberg D, St-Louis-Deschênes M (2010) The effect of acute physical exercise on cognitive function during development. Psychol Sport Exerc 11:122–126
103. Caldani S, Razuk M, Septier M, Barela JA, Delorme R, Acquaviva E et al (2018) The effect of dual task on attentional performance in children with ADHD. Front Integr Neurosci 12:67
104. Fosco WD, Kofler MJ, Alderson RM, Tarle SJ, Raiker JS, Sarver DE (2019) Inhibitory control and information processing in ADHD: comparing the dual task and performance adjustment hypotheses. J Abnorm Child Psychol 47:961–974
105. Pontifex M, Saliba B, Raine L, Picchietti D, Hillman C (2013) Exercise improves behavioral, neurocognitive, and scholastic performance in children with attention-deficit/hyperactivity disorder. J Pediatr 162:543–551
106. Cerrillo-Urbina A, García-Hermoso A, Sánchez-López M, Pardo-Guijarro M, Gómez J, Martínez-Vizcaíno V (2015) The effects of physical exercise in children with attention deficit hyperactivity disorder: a systematic review and meta-analysis of randomized control trials. Child Care Health Dev 41:779–788
107. Wang YH, Gau SF, Yang LK, Chang JC, Cheong PL, Kuo HI (2024) Acute aerobic exercise at different intensities modulates inhibitory control and cortical excitability in adults with attention-deficit hyperactivity disorder (ADHD). Asian J Psychiatr 95:103993. https://doi.org/10.1016/j.ajp.2024.103993
108. Pontifex MB, McGowan AL, Chandler MC, Gwizdala KL, Parks AC, Fenn K et al (2019) A primer on investigating the after effects of acute bouts of physical activity on cognition. Psychol Sport Exerc 40:1–22
109. Basso JC, Suzuki WA (2017) The effects of acute exercise on mood, cognition, neurophysiology, and neurochemical pathways: a review. Brain Plast 2:127–52
110. Wang C-H, Chang C-C, Liang Y-M, Shih C-M, Chiu W-S, Tseng P et al (2013) Open vs. closed skill sports and the modulation of inhibitory control. PLoS ONE 8:e55773
111. Chueh T-Y, Hsieh S-S, Tsai Y-J, Yu C-L, Hung C-L, Benzing V et al (2022) Effects of a single bout of moderate-to-vigorous physical activity on executive functions in children with attention-deficit/hyperactivity disorder: a systematic review and meta-analysis. Psychol Sport Exerc 58:102097

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